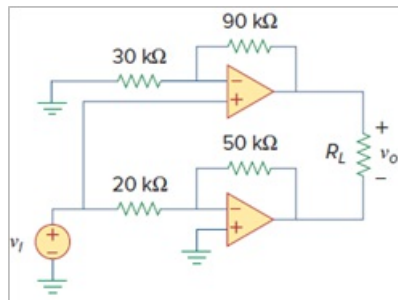


Problem

Refer to the *bridge amplifier* shown in Fig. 5.109. Determine the voltage gain v_o/v_i .

Figure 5. 109:



Step-by-step solution

Step 1 of 4

Consider the following circuit:

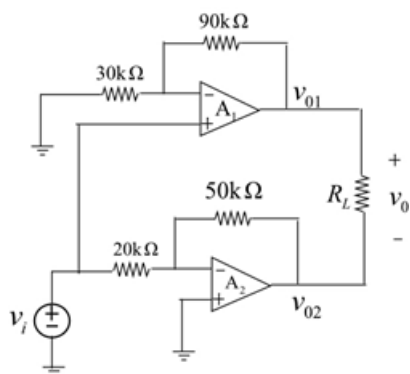


Figure 1

Step 2 of 4

Calculate the value of voltage v_{o1} of the circuit shown in figure 1.

The op-amp A_1 shown in figure 1 is a non inverting amplifier.

Therefore, the output of the op-amp A_1 is,

$$v_{o1} = \left(1 + \frac{R_f}{R} \right) v_i$$

Here, $R_f = 90 \text{ k}\Omega$ and $R = 30 \text{ k}\Omega$.

Substitute the values

$$\begin{aligned} v_{o1} &= \left(1 + \frac{90 \text{ k}\Omega}{30 \text{ k}\Omega} \right) v_i \\ &= 4 v_i \end{aligned}$$

Step 3 of 4

Calculate the value of voltage v_{02} of the circuit shown in figure 1.

The op-amp A_2 in figure 1 is an inverting amplifier.

Thus, the output of the op-amp A_2 is,

$$v_{02} = -\left(\frac{R_f}{R}\right)v_i$$

Here, $R_f = 50\text{ k}\Omega$ and $R = 20\text{ k}\Omega$.

Substitute the values.

$$\begin{aligned} v_{02} &= -\left(\frac{50\text{ k}\Omega}{20\text{ k}\Omega}\right)v_i \\ &= -2.5v_i \end{aligned}$$

Step 4 of 4

Calculate the value of voltage v_o of the circuit shown in figure 1.

Apply the Kirchhoff's voltage law.

$$v_o = v_{01} - v_{02}$$

Replace v_{01} by $4v_i$ and v_{02} by $-2.5v_i$.

$$\begin{aligned} v_o &= 4v_i - (-2.5v_i) \\ &= 6.5v_i \end{aligned}$$

$$\frac{v_o}{v_i} = 6.5$$

Therefore, the voltage gain is 6.5.